Johan Ajnabi

PhD Candidate | Wound Healing, Mechanobiology and Epigenetics

BRIC-Institute for Stem Cell Science and Regenerative Medicine (inStem)
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Summary

Mechanobiology and epigenetics researcher studying how mechanical forces regulate chromatin organization and cell fate transitions during tissue repair. My work identifies cytoskeleton-driven control of DNMT3a localization as a mechanism linking tissue mechanics to epigenetic reprogramming. I combine cell biology, signaling, and molecular biology tools to investigate how tissue architecture shapes regeneration and disease. I am interested in advancing research at the interface of mechanobiology, epithelial biology, and tissue-scale control of cell states.

Education

PhD Candidate, BRIC-Institute for Stem Cell Science and Regenerative Medicine (inStem) — *2019 - Present*
Advisor: Prof. Colin Jamora
PhD thesis: Regulation of DNMT3a localization in the cutaneous wound healing response

M.Sc. in **Molecular Biology and Biotechnology**, ICAR-Indian Agricultural Research Institute — *2019*

B.Sc. (Hons.) in **Agriculture**, Bidhan Chandra Krishi Viswavidyalaya — *2017*

Research Experience

BRIC-Institute for Stem Cell Science and Regenerative Medicine (inStem), India — *2019-Present*
Research Scholar (PhD)
Advisor: Prof. Colin Jamora

Mechanotransduction and epigenetic regulation (Primary project)
- Identified a mechanosensitive regulation mechanism of DNMT3a nucleocytoplasmic shuttling, establishing a link between cytoskeletal dynamics and epigenetic reprogramming during wound repair
- Integrated live-cell imaging, biochemical assays, and genetic perturbations to define ERK-dependent mechanotransduction pathways
- Led first-author study (bioRxiv: Ajnabi *et al.*, 2026).

Stem cell signaling and niche regulation
- Contributed to the discovery of a Mindin-integrin-STAT3 signaling axis that sustains keratinocyte stemness through integrin endocytosis (Cell Communication and Signaling: Dam *et al.*, 2026).
- Contributed to work showing how autocrine Mindin signaling maintains stem and progenitor states in skin epithelium and carcinoma cells (Cell Reports: Badarinath *et al.*, 2022).

Tissue-scale and translational biology (vitiligo, fibrosis)
- Performed spatial transcriptomic profiling of vitiligo skin before and after NB-UVB treatment and validated key findings by immunofluorescence and qPCR (Clinical & Translational Immunology: Dutta *et al.*, 2026).
- Studied extracellular matrix regulation of cutaneous fibrosis using transgenic mouse models (Journal of Investigative Dermatology: Rana *et al.*, 2023).

Host-pathogen interactions
- Investigated LL-37-mediated inhibition of SARS-CoV-2 entry using mammalian cell systems, pseudovirus assays, and flow cytometry (Frontiers in Immunology: Bhatt *et al.*, 2023).

ICAR-National Institute for Plant Biotechnology, India — *2017-2019*
M.Sc. dissertation
Advisor: Dr. Monika Dalal
- Characterized promoter cis-regulatory regions controlling *PM19* expression in wheat using cloning, transient GUS assays, plant tissue culture, transformation, and sequence analysis (Thesis: Ajnabi J, 2019)

Publications

First-Author & Lead Contributions

- **Ajnabi J**, Dam B, Gupta E, Saha T, et al. *Actin-dependent mechanotransduction controls nucleocytoplasmic partitioning of DNMT3a through ERK1/2 signaling during cutaneous wound healing.* **bioRxiv**, 2026 *(preprint)*

Collaborative Publications

- Dam B**, Ajnabi J**, et al. Mindin-mediated αM-integrin endocytosis activates STAT3 to maintain keratinocyte stemness. **Cell Communication and Signaling**, 2026
- Dutta A, Gupta D, **Ajnabi J**, et al. *Spatial transcriptomic analysis of the immune landscape following NB-UVB treatment of vitiligo skin.* **Clinical & Translational Immunology**, 2026
- Dutta S, Islam Z, Das S, Barman A, Chowdhury M, Mondal BP, **Ajnabi J**, et al. *Harmonizing plant resilience: membrane lipid dynamics in response to abiotic stresses.* **Discover Plants**, 2025
- Bhatt T, Dam B, Khedkar SU, Lall S, Pandey S, Kataria S, **Ajnabi J**, et al. *Niacinamide enhances cathelicidin-mediated SARS-CoV-2 membrane disruption.* **Frontiers in Immunology**, 2023
- Rana I, Kataria S, Tan TL, Hajam EY, Kashyap DK, Saha D, **Ajnabi J,** et al. *Mindin (SPON2) is essential for cutaneous fibrogenesis in a mouse model of systemic sclerosis.* **Journal of Investigative Dermatology**, 2023
- Badarinath K, Dam B, Kataria S, Zirmire RK, Dey R, Kansagara G, **Ajnabi J**, et al. *Snail maintains the stem/progenitor state of skin epithelial cells and carcinomas through autocrine Mindin signaling.* **Cell Reports**, 2022

Fellowships and Awards

- **Paeonia Foundation Travel Award –** Selected to present research at *Mechanobiology in Health and Disease Conference*, National University of Singapore, 2023
- **ASRB-NET (Agricultural Biotechnology)** – Qualified national eligibility test for lectureship, 2020
- **CSIR-NET Junior Research Fellowship (JRF) (AIR 20)** – National-level fellowship awarded to top-ranked candidates by the Council of Scientific and Industrial Research (CSIR), India, 2019
- **ICMR Junior Research Fellowship (JRF)** – Competitive national fellowship in Life Sciences by the Indian Council of Medical Research (ICMR), 2019
- **GATE XL (AIR 8)** – All India Rank 8 in Graduate Aptitude Test in Engineering (Life Sciences), 2019
- **DBT-BCIL Fellowship** – National-level fellowship awarded by the Department of Biotechnology, Government of India, 2019
- **ICAR AICE-JRF/SRF (AIR 3) –** All India Rank 3 in national examination for agricultural biotechnology research fellowships by the Indian Council of Agricultural Research (ICAR), 2019
- **ICAR AIEEA-PG (AIR 1)** – All India Rank 1 in national entrance examination for postgraduate studies in agricultural biotechnology, 2017

Selected Presentations

- **Poster** – "Epigenetic and mechanical regulation of the cutaneous wound healing response" at *Mechanobiology in Health and Disease Conference*, National University of Singapore, Singapore, 2023
- **Poster** – "Role of DNMT3a in the cutaneous wound healing response using a mouse model" at *10th International Conference of Laboratory Animal Scientists' Association (LASA)*, Hyderabad, India, 2022
- **Poster** – "Identification and characterization of stress-responsive PM19 promoter from wheat" at *National Agricultural Science Congress,* New Delhi, India, 2019

Skills

Experimental Expertise

- Experimental design and hypothesis-driven research
- Experimental systems: mouse models and colony management, primary mammalian cell culture
- Imaging and tissue analysis: confocal & multiphoton microscopy, Flow cytometry (FACS)
- Molecular and cell biology: genetic perturbation, transfection, qPCR, western blotting, protein purification, pseudovirus generation, BSL-2 workflows

Computational & Analytical

- GraphPad Prism, Adobe Illustrator, basic R, and Python
- Image analysis (FIJI/ImageJ)

Mentoring & Collaboration

- Mentored junior researchers in primary cell culture, molecular biology techniques, and experimental design, including training in data interpretation and scientific communication (presentations and writing)
- Collaborated on interdisciplinary projects spanning mechanobiology, immunology, and spatial transcriptomics, integrating experimental and clinical datasets
- Contributed to multi-author studies across epithelial biology, fibrosis, and host–pathogen interactions, supporting experimental design, data generation, and analysis

Languages

- Bengali (Native)
- English (Advanced)
- Hindi (Limited working proficiency)

References

Prof. Colin Jamora
Senior Professor
Department of Life Sciences, Shiv Nadar Institution of Eminence Deemed to be University, Gautam Buddha Nagar, Tehsil Dadri, Uttar Pradesh - 201314, India
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